

AUDIOSHIELD FORENSICS

Audio Forensic Analysis Report

Report ID	ASF-20260914-7A70
Generated	2026-09-14 13:44:36 India Standard Time
File analyzed	3534d565-6753-4873-848a-ccb1615eb148.mp3

This report presents separate evidence streams and conservative forensic indicators. It is not a substitute for expert examination of the original evidence.

1. Executive Summary

Overall risk assessment: **high**
Overall assessment: **likely_synthetic** | Confidence: **0.9952549573093968**

Key findings

- AI consensus: spoof (risk: high).
 - Metadata risk: low; 2 finding(s).
 - Signal risk: medium; 0 anomaly/anomalies.
- Official Eval AI comparison: CNN 69.83%, SVM 89.40%, AASIST 88.42%.
- Recommendation: Likely synthetic indicators require further investigation.

2. File Information

Property	Value
File name	3534d565-6753-4873-848a-ccb1615eb148.mp3
File size (bytes)	10417214
Format	mp3
Duration (seconds)	325.537938
Sample rate (Hz)	44100
Channels	2
Codec	mp3
Bitrate (bits/sec)	255999
Creation time	2026:09:14 13:44:31+05:30
Modification time	2026:09:14 13:44:31+05:30

3. AI Detection Results

Official Eval-set comparison

Model	Accuracy	F1	EER	Rank
CNN	69.83%	0.7978	0.0716	3rd
SVM	89.40%	0.9400	0.1698	2nd
AASIST	88.42%	0.9311	0.0638	1st

Best model: **AASIST** (best balanced rank across accuracy, F1, and EER; lower EER is better).

4. Metadata Forensics

Extracted metadata

Field	Value
File Name	3534d565-6753-4873-848a-ccb1615eb148.mp3

Field	Value
File Size	10417214
Format	mp3
Duration	325.537938
Codec	mp3
Sample Rate	44100
Channels	2
Bitrate	255999
Encoder	Not available
Creation Time	2026:09:14 13:44:31+05:30
Modification Time	2026:09:14 13:44:31+05:30
Device Info	Not available
Software	Not available
Comments	Not available

Risk score: **low**

Findings

Severity	Field	Description
low	device	Device information is missing; provenance cannot be verified from metadata alone.
low	encoder	Encoder or software information is missing.

Metadata indicators are not conclusive on their own and require further investigation.

5. Signal Forensics

Spectral features

Feature	Mean	Std	Min	Max
Spectral Centroid	1873.8022811802657	654.0664812102931	0.0	3369.1925826301203
Spectral Bandwidth	1740.0574714779364	553.3177117601309	0.0	2479.9034507143583
Spectral Rolloff	3996.3674363057326	1341.118625722523	0.0	5867.1875
Spectral Flatness	0.0851525545347694	0.2641179062152477	0.00013625748397316784	1.0000007152557373
Spectral Contrast	28.89167329882085	6.318338549561572	14.602932618194638	38.909059045828386

Temporal features

Feature	Mean	Std	Min	Max
Zero Crossing Rate	0.15166886445063693	0.06309239550361655	0.0	0.2724609375
Rms Energy	0.14173886834528826	0.06199906811914533	0.0	0.27541640400886536
Pitch	148.34764121216477	48.45034515667105	65.40639132514966	195.99771799087463

Risk score: **medium**

Anomalies

No signal-level anomalies were reported.

6. Evidence Fusion

Evidence source	Risk / result
AI	spoof (risk: high)
Metadata	low
Signal	medium
Overall	high

Assessment: **likely_synthetic** | Confidence: **0.9952549573093968**

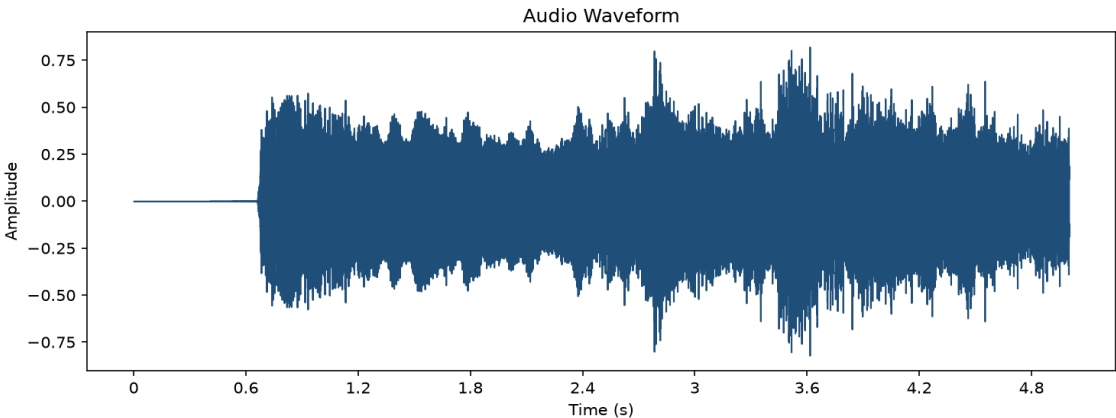
Reasoning: Source risks are AI=high, metadata=low, signal=medium; applied documented rule-based logic.

Recommendation: Likely synthetic indicators require further investigation.

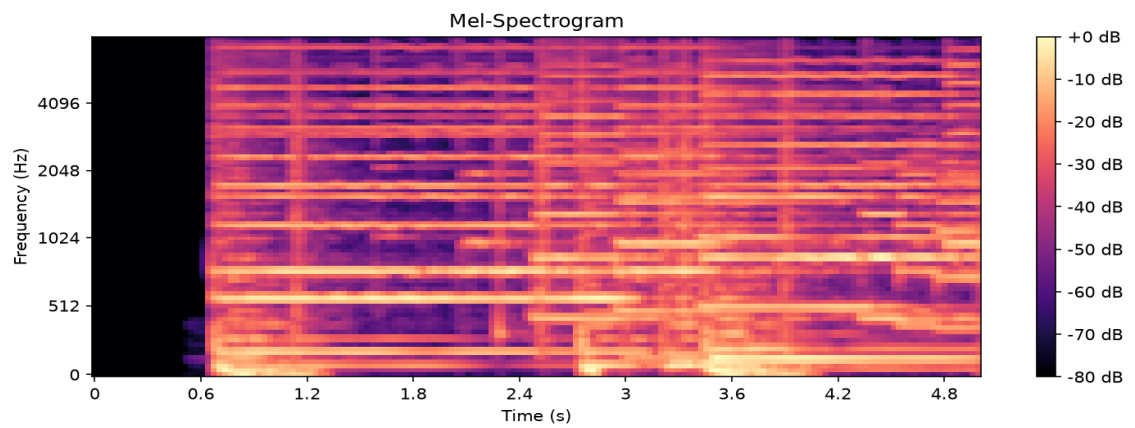
AI model breakdown

Model	Label	Confidence
CNN	spoof	1.0
SVM	spoof	0.9947267878980394
AASIST	spoof	0.9910380840301514

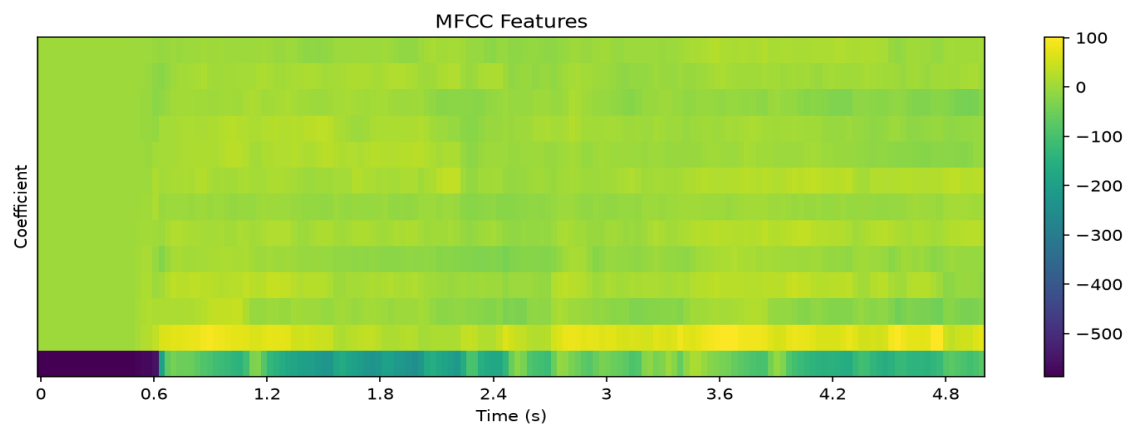
7. Visual Analysis



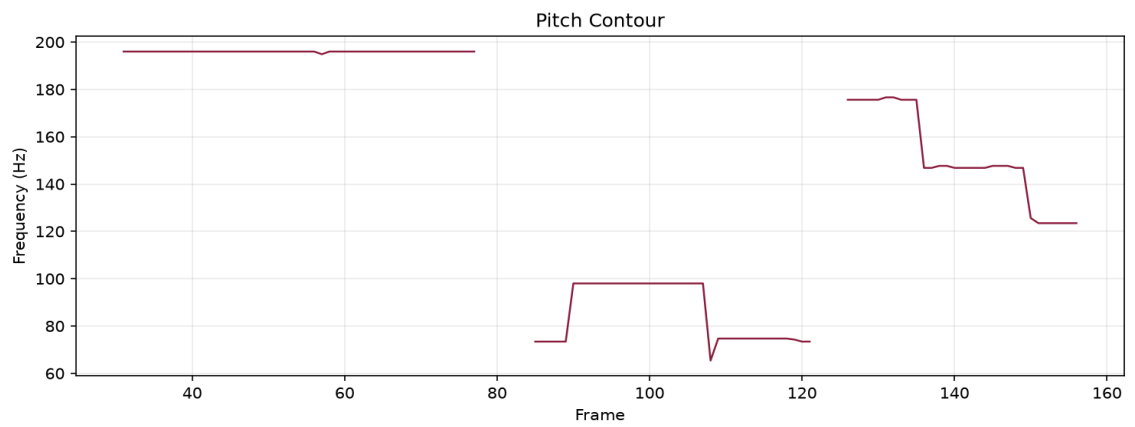
Waveform: Time-domain waveform.



Spectrogram: Frequency content over time.



MFCC: Mel-frequency cepstral coefficients.



Pitch contour: Estimated fundamental frequency.

8. Conclusion

The combined evidence indicates an overall assessment of **likely_synthetic** with a reported risk level of **high**. These are indicators from the available reports and should be interpreted with the original audio and chain of custody.

Limitations: AI results may be unavailable; metadata can be edited or lost; signal features are influenced by recording conditions and preprocessing; and this report does not establish authenticity by itself.

9. Appendix

Raw AI evaluation results

```
{ "CNN": { "model": "CNN", "checkpoint":
"C:\\Users\\allen\\AudioShield-Forensics\\models\\cnn\\cnn_final_best.pth", "dataset": "ASVspoof 2019 LA",
"split": "eval", "total_samples": 71237, "bonafide_samples": 7355, "spoof_samples": 63882, "best_epoch": 4,
"best_validation_loss": 0.003000220499216719, "threshold": { "value": 0.40700000000000003, "source":
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"selection_source": "dev_only", "path":
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"min-max", "train_min": -80.0, "train_max": 0.0, "source":
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0.7977719441856964, "specificity": 0.9993201903467029, "fpr": 0.0006798096532970768, "tpr":
0.6636298174759713, "balanced_accuracy": 0.8314750039113371, "threshold": 0.40700000000000003,
"selection_method": "balanced_accuracy_with_spoof_recall_constraint" }, "confusion_matrix": { "labels": [
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"official_eval", "total_samples": 71237, "bonafide_samples": 7355, "spoof_samples": 63882, "metrics": {
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0.9400100102485879, "roc_auc": 0.874909277304299, "eer": 0.16983042600362058 }, "AASIST": {
"total_samples": 71237, "bonafide_samples": 7355, "spoof_samples": 63882, "tn": 7221, "fp": 134, "fn": 8115,
"tp": 55767, "accuracy": 0.8842034336089392, "precision": 0.9976029051358651, "recall": 0.8729689114304499,
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0.9722707898317903 } }
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Raw metadata report

```
{ "metadata": { "file_name": "3534d565-6753-4873-848a-ccb1615eb148.mp3", "file_size": "10417214", "format":
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"encoder": null, "creation_time": "2026:09:14 13:44:31+05:30", "modification_time": "2026:09:14
13:44:31+05:30", "device_info": null, "software": null, "comments": null, "raw_ffprobe": { "streams": [ {
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"codec_tag_string": "[0][0][0][0]", "codec_tag": "0x0000", "mime_codec_string": "mp4a.40.34", "sample_fmt":
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"initial_padding": 0, "r_frame_rate": "0/0", "avg_frame_rate": "0/0", "time_base": "1/14112000", "start_pts":
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"disposition": { "default": 0, "dub": 0, "original": 0, "comment": 0, "lyrics": 0, "karaoke": 0, "forced": 0,
"hearing_impaired": 0, "visual_impaired": 0, "clean_effects": 0, "attached_pic": 0, "timed_thumbnails": 0,
"non_diegetic": 0, "captions": 0, "descriptions": 0, "metadata": 0, "dependent": 0, "still_image": 0,
"multilayer": 0 } } ], "format": { "filename":
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"probe_score": 52 } }, "raw_exiftool": { "SourceFile":
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13.59, "FileName": "3534d565-6753-4873-848a-ccb1615eb148.mp3", "Directory":
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13:44:31+05:30", "FileAccessDate": "2026:09:14 13:44:34+05:30", "FileCreateDate": "2026:09:14
13:44:31+05:30", "FilePermissions": "-rw-rw-rw-", "FileType": "MP3", "FileTypeExtension": "mp3", "MIMEType":
"audio/mpeg", "MPEGAudioVersion": 1, "AudioLayer": 3, "AudioBitrate": "256 kbps", "SampleRate": 44100,
"ChannelMode": "Joint Stereo", "MSStereo": "On", "IntensityStereo": "Off", "CopyrightFlag": true,
"OriginalMedia": true, "Emphasis": "None", "Duration": "0:05:26 (approx)" } }, "summary": { "file_name":
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"creation_time": "2026:09:14 13:44:31+05:30", "modification_time": "2026:09:14 13:44:31+05:30",
"device_info": null, "software": null, "comments": null }, "analysis": { "findings": [ { "field": "device",
"has_issue": true, "severity": "low", "description": "Device information is missing; provenance cannot be
verified from metadata alone." }, { "field": "encoder", "has_issue": true, "severity": "low", "description":
"Encoder or software information is missing." } ], "risk_score": "low", "suspicious_fields": [],
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Raw signal report

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{ "spectral_features": { "spectral_centroid": { "mean": 1873.8022811802657, "std": 654.0664812102931, "min":
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553.3177117601309, "min": 0.0, "max": 2479.9034507143583 }, "spectral_rolloff": { "mean": 3996.3674363057326,
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"std": 0.2641179062152477, "min": 0.00013625748397316784, "max": 1.0000007152557373 }, "spectral_contrast": {
"mean": 28.89167329882085, "std": 6.318338549561572, "min": 14.602932618194638, "max": 38.909059045828386 }
}, "temporal_features": { "zero_crossing_rate": { "mean": 0.15166886445063693, "std": 0.06309239550361655,
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"min": 0.0, "max": 0.27541640400886536 }, "pitch": { "mean": 148.34764121216477, "std": 48.45034515667105,
"min": 65.40639132514966, "max": 195.99771799087463 }, "anomalies": { "anomalies": [], "risk_score": "low",
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changes; requires further investigation." }, "energy_analysis": { "energy_mean": 0.1417388617992401,
"energy_std": 0.06199906766414642, "anomalies": [] } }, "plots": [ "results\\signal_plots\\waveform.png",
"results\\signal_plots\\spectrogram.png", "results\\signal_plots\\mfcc.png",
"results\\signal_plots\\pitch_contour.png" ], "signal_risk": "medium", "summary": "Signal risk is medium. No
broad signal-level anomalies identified." }
```

Methodology

This report compiles the supplied AI, metadata, signal, and fusion JSON outputs. No new model inference is performed during report generation. Evidence sources remain separated to preserve interpretability.